Suggested problems: 1, 3, 6, 8

Problem 1: Consider the follow p.d.f. on the real line:

$$f(x) = \begin{cases} cx^5 e^{-x} & : x > 0, \\ 0 & : \text{else}, \end{cases}$$

for some constant c.

(a) Compute the constant c.

(b) Let $X_1, \ldots, X_n$ be the heights (in feet) of a random sample of people chosen from our school. Explain why it is reasonable to believe that these random variables are independent.

(c) Explain why $f(x)$ might be a reasonable p.d.f. for the X_i's.

(d) Suppose $n \geq 2$. What is the probability that the heights of the first and second people are both less than six feet?

Problem 2: Suppose a fair six-sided die is rolled n times. Let X_i be the result of the i-th roll.

(a) Explain why $X_1, \ldots, X_n$ is an i.i.d. sequence of random variables. What is their common distribution?

(b) Compute the probability that each roll results in a 3.